

Oka complements in algebraic tori and their logarithmic covers

Yun-Heng Du

Bin Guo

Peng-Chao Wang

Song-Yan Xie

October 11, 2026

Abstract

Suppose $n \geq 2$ and $K \subset \mathbb{C}^n$ is compact and polynomially convex. Write $q(z) = (e^{-iz_1}, \dots, e^{-iz_n})$, $X_K = (\mathbb{C}^*)^n \setminus (K \cap (\mathbb{C}^*)^n)$, and $Y_K = \mathbb{C}^n \setminus q^{-1}(K)$. We prove that X_K and its logarithmic covering Y_K are Oka. More strongly, Y_K admits normalized holomorphic families of entire embeddings through prescribed holomorphic sections. The proof combines polynomial pair Hamiltonian transport in multiplicative coordinates, compatible with the coordinate hyperplanes, with a fixed Fatou–Bieberbach model. Applications include Oka slab crosses over arbitrary nonempty connected planar domains, complements of periodic equal-radius Euclidean balls centered at products of planar lattices, and complements of sufficiently small closed metric balls in products of elliptic curves.

2020 Mathematics Subject Classification. Primary 32Q56; Secondary 32E20, 32E30, 32H02.

1 Introduction

Kusakabe [12, Question 5.6] asks whether equal-radius balls centered at an arbitrary discrete set $D \subset \mathbb{C}^n$, $n > 1$, have an Oka complement when the infimum of the distances between distinct centers is greater than twice the radius. His Corollary 5.7 covers the case $D = \mathbb{Z} \times \{0\}^{n-1}$, and Theorem 1.3 below treats the case in which D is a full product of planar lattices. Forstnerič [5, p. 23, paragraph after Proposition 6.1.4] asked whether removing sufficiently small balls from a complex torus preserves the Oka property, following earlier results on dominability; for products of elliptic curves with their flat product metrics, Corollary 1.4 answers this for one ball.

The two questions are addressed by one construction. For broad introductions to modern Oka theory, including its geometric and homotopy-theoretic viewpoints, see Forstnerič [7], Forstnerič–Lárusson [8], and Lárusson [13]. Recall that a complex manifold is Oka if it has the convex approximation property: for every $m \geq 1$, every holomorphic map from a neighborhood of a compact convex set in $\mathbb{C}^m$ can be approximated uniformly on that set by holomorphic maps from $\mathbb{C}^m$. We use Kusakabe’s equivalent convex spray criterion [11, Theorem 2.2] and his localization theorem [11, Theorem 1.4]; the latter applies to analytic–Zariski open neighborhoods, that is, to complements of closed complex subvarieties.

The closest published antecedent for the parameter-dependent mechanism used here is the work of Forstnerič and Fornæss Wold. They constructed holomorphic families of Fatou–Bieberbach domains avoiding compact holomorphically convex obstacles, with a section-transport step by fiber automorphisms that remain close to the identity near the obstacle [9, Theorems 1.1 and 3.1; Lemma 3.2]. Their subsequent work treats a compact polynomially convex obstacle together with one coordinate hyperplane [10, Theorem 2.3 and Corollary 3.2]. These constructions provide the transport-and-fixed-model pattern that we adapt below.

In our earlier work [3], complements of tubes around the real plane and products of closed annuli in $\mathbb{C}^2$ were shown to be Oka by constructing holomorphic families of embeddings through prescribed holomorphic sections. In particular, [3, Proposition 1.2, version 1] gives a normalized holomorphic family of Fatou–Bieberbach maps outside a two-dimensional tube, together with a holomorphic family

of attracting automorphisms. The logarithmic obstacles below are unbounded and may interact with several coordinate hyperplanes; controlling their full logarithmic preimages requires the new displacement argument developed here.

We regard $(\mathbb{C}^*)^n$ as the n -dimensional complex algebraic torus. For $K \subset \mathbb{C}^n$, write

$$\begin{aligned} q(z) &= (e^{-iz_1}, \dots, e^{-iz_n}), & E_K &= q^{-1}(K), \\ Y_K &= \mathbb{C}^n \setminus E_K, & X_K &= (\mathbb{C}^*)^n \setminus (K \cap (\mathbb{C}^*)^n). \end{aligned}$$

The restriction $q : Y_K \rightarrow X_K$ is the logarithmic covering associated with K . The following theorem is the technical core of the paper.

Theorem 1.1. *For $n \geq 2$ and compact polynomially convex $K \subset \mathbb{C}^n$, the following statements hold.*

- (i) *The manifolds Y_K and X_K are nonempty, path connected, and Oka.*
- (ii) *For every compact convex $L_0 \subset \mathbb{C}^m$ and holomorphic map $f : U_0 \rightarrow Y_K$ from a neighborhood of L_0 , there are a neighborhood $V \subset U_0$ of L_0 and a holomorphic map*

$$F : V \times \mathbb{C}^n \longrightarrow Y_K, \quad F(\zeta, 0) = f(\zeta), \quad D_\xi F(\zeta, 0) = I_n,$$

such that every fiber map $F(\zeta, \cdot)$ is biholomorphic onto its image.

Part (ii) is stronger than the Oka conclusion: locally in the parameter, it gives normalized entire embeddings through a prescribed section with prescribed first jet. If $E_K \neq \emptyset$, their images are Fatou–Bieberbach domains in $\mathbb{C}^n$ that avoid E_K ; if $E_K = \emptyset$, the construction is a translation with image $\mathbb{C}^n$. This is the form used with Kusakabe’s convex spray criterion, and it is the reusable engine behind all the applications below.

Theorem 1.1 extends the normalized family construction of [3, Proposition 1.2, version 1] from two-dimensional tubes to the logarithmic obstacles $E_K \subset \mathbb{C}^n$. The compact-quotient construction of [3, Theorem 1.1, version 2] assumes a compact holomorphically convex obstacle in $(\mathbb{C}^*)^n$. Relative to the compact-obstacle families in [9] and the one-coordinate-hyperplane setting in [10], the additional issue here is that K may meet several coordinate hyperplanes. Thus $K \cap (\mathbb{C}^*)^n$ can be noncompact and the imaginary parts of E_K need not be bounded in both directions. This is the situation for periodic balls: the closure of their coordinatewise inverted obstacle has boundary strata on the coordinate hyperplanes, and Lemma 6.1 shows that this closure is polynomially convex.

The proof has three layers. First, polynomial pair Hamiltonian potentials in multiplicative coordinates produce complete flows that extend across the coordinate hyperplanes. Second, an accumulated displacement variable converts uniform control on the unbounded logarithmic obstacle E_K into approximation on a compact neighborhood of K . Third, the resulting automorphisms transport a prescribed section to a fixed Fatou–Bieberbach model; pulling the model back and normalizing the derivative gives the family in Theorem 1.1. Kusakabe’s convex spray criterion then yields the Oka property, while the logarithmic covering transfers it to X_K .

For nonempty connected planar domains $U_1, \dots, U_n$, the slab cross is

$$\mathcal{S}(U_1, \dots, U_n) = \bigcup_{j=1}^n \mathbb{C}^{j-1} \times U_j \times \mathbb{C}^{n-j}.$$

Every point $z \in \mathcal{S}(U_1, \dots, U_n)$ lies in a spanning tube contained in the cross. Indeed, choose i with $z_i \in U_i$, choose $k \neq i$ and $b \in z + \mathbb{C}e_k$ with $b_k \in U_k$, and use the coordinate lines $z + \mathbb{C}e_j$ for $j \neq i$ together with $b + \mathbb{C}e_i$. They form a connected spanning tree whose sufficiently small tubular neighborhood remains in the cross. Hence Alarcón and Forstnerič [1, Corollary 2.4] imply that every slab cross is Oka-1. The following theorem strengthens this conclusion to the full Oka property.

Theorem 1.2. *For every $n \geq 2$ and nonempty connected open sets $U_1, \dots, U_n \subset \mathbb{C}$, the slab cross $\mathcal{S}(U_1, \dots, U_n)$ is Oka.*

The strength of the statement is that no boundedness or boundary regularity of the domains is assumed. When $0 \in U_j$ for every j , the cross contains a neighborhood of each coordinate hyperplane; in particular,

$$\Omega_n = \{z \in \mathbb{C}^n : \min_{1 \leq j \leq n} |z_j| < 1\}$$

is Oka. For $n = 2$, it is $(\Delta \times \mathbb{C}) \cup (\mathbb{C} \times \Delta)$, where Δ denotes the open unit disc.

The second application is to periodic Euclidean balls: for $c \in \mathbb{C}^d$ and $r > 0$, the closed Euclidean ball of radius r about c is

$$\overline{B}(c, r) = \{z \in \mathbb{C}^d : |z - c| \leq r\}.$$

For rank-two planar lattices $\Lambda_j \subset \mathbb{C}$, $1 \leq j \leq n$, write

$$\Lambda = \Lambda_1 \times \dots \times \Lambda_n, \quad \ell_j = \min_{0 \neq \lambda \in \Lambda_j} |\lambda|, \quad \ell = \min_j \ell_j.$$

With respect to the product Euclidean norm, ℓ is the length of a shortest nonzero vector of Λ .

Theorem 1.3. *If $n \geq 2$ and $0 < 2r < \ell$, then*

$$\Omega = \mathbb{C}^n \setminus \bigcup_{\lambda \in \Lambda} \overline{B}(\lambda, r)$$

is Oka.

The substantive new step is the compactification of the infinite periodic obstacle: after coordinatewise inversion, Lemma 6.1 identifies a compact polynomially convex closure with explicit boundary strata.

Equip

$$T = (\mathbb{C}/\Lambda_1) \times \dots \times (\mathbb{C}/\Lambda_n)$$

with the flat product metric induced by the Euclidean norm on $\mathbb{C}^n$. Its injectivity radius is $\ell/2$, and we write $\overline{B}_T(c, r)$ for its closed metric balls.

Corollary 1.4. *For $n \geq 2$, every $c \in T$, and $0 < r < \ell/2$, the complement of the closed metric ball $\overline{B}_T(c, r)$ in T is Oka.*

This is the quotient consequence of Theorem 1.3: the injectivity radius condition makes the full inverse image of the ball exactly the periodic ball obstacle.

Translated inversion charts reduce both applications to Theorem 1.1, with Kusakabe's localization theorem handling the change of chart. For slab crosses the inverted obstacle is a product of compact planar sets. For periodic balls the coordinate boundary strata are coordinatewise inversions of balls of lower dimension, and it remains to show that the whole compactified obstacle is polynomially convex (Lemma 6.1).

Section 2 contains the connectedness statement and the approximation lemmas, Sections 3 and 4 the transport argument and the proof of Theorem 1.1, and Sections 5–7 the applications. The fixed two-dimensional Fatou–Bieberbach model is constructed in Appendix A.

2 Connectedness and polynomial approximation

This section separates the preliminary tasks used in Theorem 1.1. Lemma 2.1 supplies the topology of the base and its logarithmic covering, while the remaining lemmas provide the convexity, approximation, and covering tools used by the transport construction and its applications. For background on polynomial convexity and approximation, see [15].

Lemma 2.1. *For compact polynomially convex $K \subset \mathbb{C}^n$, both X_K and Y_K are nonempty and path connected.*

Proof. Choose $R > 0$ with $|w_j| < R$ for every $w \in K$ and every j , and set $b = (R, \dots, R) \in X_K$. Since q is surjective, $Y_K \neq \emptyset$ as well. The complement $M = \mathbb{C}^n \setminus K$ is connected: any component other than the one containing the exterior of a large ball would be bounded, with boundary in K , and the maximum-modulus principle would force it into $\widehat{K} = K$. Thus M is a connected open set, hence path connected.

Inside a ball in M , two points outside the hyperplanes can be joined in their complex affine line, whose intersection with the hyperplanes consists of at most n points. A finite chain of such balls along a path in M , with joining points chosen outside the hyperplanes, gives a path in X_K , so X_K is path connected as well.

The loops

$$\gamma_j(t) = (R, \dots, Re^{2\pi it}, \dots, R), \quad 0 \leq t \leq 1, \quad 1 \leq j \leq n,$$

lie in X_K . Concatenations of their lifts and reversed lifts join every point of $q^{-1}(b) = b' + 2\pi\mathbb{Z}^n$ to a fixed $b' \in q^{-1}(b)$. Any $z \in Y_K$ can first be joined to this fiber by lifting a path in X_K from $q(z)$ to b , and then to b' . Thus Y_K is path connected. $\square$

The next lemma is the gluing device for the two local prescriptions in Lemma 3.2: one is imposed near an ambient compact set and the other along a disjoint closed submanifold. It is the disjoint-submanifold case of Forstnerič's convexity lemma [4, Lemma 6.5]; the Stein form needed here is proved directly.

Lemma 2.2. *Let Z be Stein, $G \subset Z$ a closed complex submanifold, $P \subset Z \setminus G$ compact and $\mathcal{O}(Z)$ -convex, and $Q \subset G$ compact and $\mathcal{O}(G)$ -convex. Then $P \cup Q$ is $\mathcal{O}(Z)$ -convex.*

Proof. Suppose first that $x \notin P \cup Q$ and $x \notin G$. By Cartan's theorem A there is $h_0 \in \mathcal{O}(Z)$ that vanishes on G but not at x , and by the $\mathcal{O}(Z)$ -convexity of P there is $a \in \mathcal{O}(Z)$ with $|a(x)| > \sup_P |a|$. For large N the function $h_0 a^N$ then separates x from $P \cup Q$: it vanishes on Q , while its modulus at x exceeds its supremum on P .

Suppose next that $x \in G \setminus Q$. Choose $g \in \mathcal{O}(G)$ with $|g(x)| > \sup_Q |g|$, so that in particular $g(x) \neq 0$, and extend g to $\tilde{g} \in \mathcal{O}(Z)$ with $\tilde{g}(x) = g(x)$ by Cartan's theorem B. We claim that there is $h \in \mathcal{O}(Z)$ with $h|_G = 1$ and $\sup_P |h| < 1$; to construct it, we use Cartan's theorem A together with the compactness of P to find finitely many global functions h_j vanishing on G with no common zero near P . A Stein neighborhood W of P on which they have no common zero admits a solution $\sum h_j b_j = 1$ by Cartan's theorem B. Approximating the b_j on P by global functions B_j , which is possible by Oka–Weil, we obtain $h = 1 - \sum h_j B_j$ with the required properties. Now $\tilde{g}h^N$ agrees with g on Q and takes the value $g(x)$ at x , whereas

$$\sup_P |\tilde{g}h^N| \leq \left(\sup_P |\tilde{g}|\right) \left(\sup_P |h|\right)^N \longrightarrow 0.$$

For N large enough, its modulus at x therefore exceeds both its supremum on P and its supremum on Q . $\square$

The transport argument requires approximation not only of the local potentials but also of the fiber derivatives that determine their Hamiltonian fields. The following fixed-neighborhood form makes that simultaneous control explicit.

Lemma 2.3. *Suppose B is Stein and $S \subset B \times \mathbb{C}^n$ is compact and $\mathcal{O}(B \times \mathbb{C}^n)$ -convex. A function holomorphic near S can be approximated uniformly, together with any fixed finite number of fiber*

derivatives, on a compact neighborhood of S chosen inside its domain and fixed independently of the approximation error, by polynomials

$$P(b, w) = \sum_{k \in A} c_k(b) w^k, \quad A \subset \mathbb{Z}_{\geq 0}^n \text{ finite}, \quad c_k \in \mathcal{O}(B).$$

Proof. Choose, and then fix, nested compact $\mathcal{O}(B \times \mathbb{C}^n)$ -convex neighborhoods of S inside the domain of the given function. Oka–Weil approximation on the larger one produces a function in $\mathcal{O}(B \times \mathbb{C}^n)$, and Cauchy’s estimates then control its derivatives on the smaller one. Expanding this approximant in a Taylor series in w about zero, we obtain coefficients holomorphic on B , and the finite truncations converge together with all fiber derivatives uniformly on compact subsets. $\square$

The final preliminary result is the bridge from a convenient covering space to the target manifold. It will be used for $Y_K \rightarrow X_K$, for product holes in elliptic curves, and for complements of balls in flat product tori.

Lemma 2.4. *Suppose $p : Y \rightarrow X$ is an unramified holomorphic covering of connected complex manifolds. If Y is Oka, then X is Oka.*

Proof. This is the covering invariance of the Oka property; see [6, Proposition 2.8]. $\square$

3 Polynomial transport

We approximate transport by compositions of complete holomorphic flows, as in Andersén–Lempert theory [2]; here the polynomial exponential modes must also extend across the coordinate hyperplanes. The first lemma preserves global automorphisms while discretizing a nonautonomous flow; the second uses that mechanism to move a section while remaining uniformly close to the identity on the possibly unbounded logarithmic obstacle.

For $1 \leq r < s \leq n$ and a scalar potential H in additive coordinates, the pair Hamiltonian field is

$$V_{rs,H} = (\partial_{z_s} H) e_r - (\partial_{z_r} H) e_s.$$

If $H(z) = c q(z)^k$, where $k \in \mathbb{Z}_{\geq 0}^n$, its velocity is

$$v_{rs,k}(w) = icw^k(-k_s e_r + k_r e_s), \quad w = q(z). \quad (1)$$

Since $k \cdot v_{rs,k} = 0$, the flow is complete and is given by the formula

$$z \mapsto z + \tau v_{rs,k}(q(z)), \quad \tau \in \mathbb{C}. \quad (2)$$

The identity $q(z + \tau v_{rs,k}(q(z)))^k = q(z)^k$ establishes both the flow law and the inverse, the latter obtained by replacing τ with $-\tau$. A mode with $k_r = k_s = 0$ vanishes identically.

We introduce an accumulated displacement $h \in \mathbb{C}^n$. The corresponding augmented field on $\mathbb{C}_w^n \times \mathbb{C}_h^n$ is

$$\tilde{V}_{rs,k}(w, h) = (-i \operatorname{diag}(w) v_{rs,k}(w), v_{rs,k}(w)). \quad (3)$$

Its complete entire flow is

$$(w, h) \mapsto \left((w_j e^{-i\tau v_{rs,k,j}(w)})_{j=1}^n, h + \tau v_{rs,k}(w) \right). \quad (4)$$

Since w^k is invariant along this flow, the velocity is constant. Because the exponents are nonnegative, both formulas are entire on all of $\mathbb{C}_w^n \times \mathbb{C}_h^n$, including the coordinate hyperplanes.

If a composition starts at $(q(z), 0)$, induction over its factors gives the identity

$$w_{\text{final}} = q(A(z)), \quad h_{\text{final}} = A(z) - z, \quad (5)$$

where A is the corresponding composition of additive automorphisms (2). If c depends holomorphically on further parameters, then the automorphisms and their inverses form jointly holomorphic families in those parameters.

The completeness of the frozen fields is decisive in the following approximation: every approximating factor remains a global automorphism, rather than merely a local flow map.

Lemma 3.1. *Suppose finitely many entire vector fields $W_\nu(t, \zeta, y)$ depend holomorphically on ζ , are continuously differentiable in real $t \in [0, 1]$, and are complete for frozen (t, ζ) . Assume that the trajectories of $\dot{y} = \sum_\nu W_\nu(t, \zeta, y)$ starting from a compact family of initial data, with parameters in a compact set, exist through time one and stay in a compact set. Then the time-one map of this equation is uniformly approximable by finite compositions of flows of the frozen fields, and these compositions and their inverses are holomorphic in (ζ, y) .*

Proof. Choose a compact neighborhood of all the exact trajectories, together with a positive buffer around it. The fields, their first spatial derivatives, and their time derivatives are uniformly bounded on this neighborhood. Writing $a = 1/N$ and using the successive frozen flows at $t_j = j/N$, we obtain

$$y \mapsto y + a \sum_\nu W_\nu(t_j, \zeta, y) + O(a^2).$$

The exact nonautonomous step admits the same expansion, uniformly on the buffer. The accumulated error therefore satisfies $e_{j+1} \leq (1 + Ca)e_j + Ca^2$ with $e_0 = 0$, whence $e_j \leq C'a$ for $j \leq N$. Every factor is defined globally, since the frozen fields are complete, and its dependence on (ζ, y) , like that of its inverse, is holomorphic. For large N the accumulated error keeps all intermediate steps inside the buffer, and the approximation follows. $\square$

The next lemma is the conceptual core of the construction. Its two estimates encode competing requirements: move the prescribed section along its path, but move every point of E_K by arbitrarily little. Multiplicative coordinates and the accumulated displacement turn the latter, noncompact condition into control near the compact set K .

Lemma 3.2. *Suppose $n \geq 2$, $K \subset \mathbb{C}^n$ is compact and polynomially convex, $U \subset \mathbb{C}^m$ is convex and open, and $L \Subset U$ is compact and convex. If $p : \mathbb{C} \times U \rightarrow \mathbb{C}^n$ is holomorphic and*

$$p(t, \zeta) \notin E_K \quad (0 \leq t \leq 1, \zeta \in L).$$

For every $\epsilon > 0$ there is a holomorphic family of automorphisms A_ζ of $\mathbb{C}^n$, $\zeta \in U$, with jointly holomorphic inverse and

$$\sup_{\zeta \in L, z \in E_K} |A_\zeta(z) - z| < \epsilon, \quad (6)$$

$$\sup_{\zeta \in L} |A_\zeta(p(0, \zeta)) - p(1, \zeta)| < \epsilon. \quad (7)$$

Proof. If L is empty, the identity family suffices; if K is empty, so does the translation by $p(1, \zeta) - p(0, \zeta)$. In the remaining case, set $B = \mathbb{C} \times U$ and $J = [0, 1] \times L$. The proof has four stages. We first prescribe local Hamiltonian potentials near the obstacle and the section. We then approximate them by polynomials with derivative control. The augmented flow converts compact control near K into displacement control on all of E_K , and a final discretization by frozen complete flows produces the required automorphisms.

Step 1: local Hamiltonian data. In $Z = B \times \mathbb{C}_w^n$, consider

$$P_0 = J \times K, \quad G = \{(t, \zeta, w) : w = q(p(t, \zeta))\}, \quad Q_0 = G|_J.$$

The product P_0 is $\mathcal{O}(Z)$ -convex, because J is convex and K is polynomially convex. The closed graph G is biholomorphic to B , and Q_0 is $\mathcal{O}(G)$ -convex. Avoidance gives $P_0 \cap G = \emptyset$, so Lemma 2.2 makes $P_0 \cup Q_0$ $\mathcal{O}(Z)$ -convex.

Set $v = \partial_t p$ and $a = q(p)$. The disjoint compact sets P_0 and Q_0 have positive distance, so we may choose disjoint open neighborhoods N_P of P_0 and N_Q of Q_0 . Each a_j is nonzero, and the open set

$$\left\{ (t, \zeta, w) : \max_j \left| \frac{w_j}{a_j(t, \zeta)} - 1 \right| < \frac{1}{2} \right\}$$

contains Q_0 . Intersecting N_Q with this set leaves it disjoint from N_P and still containing Q_0 , while every ratio now lies in the fixed disc $|\xi - 1| < 1/2$. On that disc we use the holomorphic logarithm normalized by $\log 1 = 0$. The $n - 1$ potentials vanish on N_P and have the following formulas on N_Q :

$$H_{12}(t, \zeta, w) = i \left(v_1 \log \frac{w_2}{a_2} - v_2 \log \frac{w_1}{a_1} \right), \quad (8)$$

$$H_{1j}(t, \zeta, w) = -iv_j \log \frac{w_1}{a_1}, \quad 3 \leq j \leq n. \quad (9)$$

These prescriptions give holomorphic potentials on the disjoint union $N_P \cup N_Q$. On a sufficiently small additive tube about $p(t, \zeta)$, their pullbacks by q are

$$v_1(z_2 - p_2) - v_2(z_1 - p_1), \quad -v_j(z_1 - p_1),$$

and their pair Hamiltonian fields sum to the constant translation v in all n coordinates.

Step 2: polynomial approximation with derivative control.

Fix compact $\mathcal{O}(Z)$ -convex neighborhoods S_-, S_+ of $P_0 \cup Q_0$ with $S_- \subset \text{int } S_+ \subset S_+ \subset N_P \cup N_Q$. Each $S_\pm$ is the union of the disjoint compact parts $S_\pm \cap N_P$ and $S_\pm \cap N_Q$. Fix a bounded neighborhood N of K with $J \times \overline{N} \subset \text{int } S_- \cap N_P$, and set

$$d = \text{dist}(K, \mathbb{C}^n \setminus N) > 0, \quad C_N = \max_{w \in \overline{N}} \max_j |w_j|.$$

Choose $\eta > 0$ with $C_N \eta < d$ and $\eta < \epsilon/2$.

Each potential is holomorphic on an open neighborhood of the whole S_+ , so Lemma 2.3 applies on S_+ and approximates all the potentials simultaneously. For each potential H and its polynomial approximant $\hat{H}$, Cauchy's estimates give

$$\sup_{S_-} |\partial_w^\alpha (\hat{H} - H)| \leq C_\alpha \sup_{S_+} |\hat{H} - H|$$

for the required fiber derivatives, with constants determined by the fixed neighborhoods and their positive buffer. The sum of the approximating pair Hamiltonian fields has velocity $\mathcal{V}(t, \zeta, w)$. The local potentials vanish identically on $J \times \overline{N}$, and their fiber derivatives vanish there as well; hence the Cauchy bound together with $|w_j| \leq C_N$ in the Euler operators $-iw_j \partial_{w_j}$ gives $|\mathcal{V}| < \eta$ on $J \times \overline{N}$, including where some coordinates of w vanish, once the errors $\sup_{S_+} |\hat{H} - H|$ are sufficiently small.

The approximating potentials have no negative powers, so their pullbacks and vector fields are entire and single-valued in z . On the tube about the section, the local logarithms have the affine pullbacks displayed above. Fix $\delta > 0$, before choosing the approximation error, so that

$$(t, \zeta, q(p(t, \zeta) + u)) \in \text{int } S_- \cap N_Q \quad ((t, \zeta) \in J, |u| \leq \delta).$$

Choose $0 < \theta < \min\{\delta/2, \epsilon/2\}$ and make the derivative approximation close enough that

$$|\mathcal{V}(t, \zeta, q(p(t, \zeta) + u)) - v(t, \zeta)| < \theta$$

throughout this fixed tube.

Step 3: compact control in the augmented system.

Consider the nonautonomous augmented system

$$\dot{w} = -i \operatorname{diag}(w) \mathcal{V}(t, \zeta, w), \quad \dot{h} = \mathcal{V}(t, \zeta, w). \quad (10)$$

For initial data $(w(0), h(0)) = (w_0, 0)$ with $w_0 \in K$, the system gives, as long as $w(t)$ has not left N ,

$$|h(t)| \leq \eta t, \quad |w(t) - w_0| \leq C_N \eta t.$$

The second inequality prevents a first exit before time one. Since $C_N \eta < d$, the trajectories with initial data in K therefore remain in the compact set

$$\{w : \operatorname{dist}(w, K) \leq C_N \eta\} \times \{h : |h| \leq \eta\} \subset N \times \mathbb{C}^n,$$

and $|h(1)| < \epsilon/2$, uniformly in $\zeta \in L$, including for initial points of K on the coordinate hyperplanes.

For the section initial data $(q(p(0, \zeta)), 0)$, the component equations in (10) give $\frac{d}{dt}(w_j(t)e^{ih_j(t)}) = 0$. Since $h(0) = 0$, this yields

$$w(t) = q(p(0, \zeta) + h(t)).$$

Set $e(t) = p(0, \zeta) + h(t) - p(t, \zeta)$, so that $e(0) = 0$. On the maximal forward interval of existence, while $t \leq 1$ and $|e(t)| < \delta$, the tube estimate gives $|\dot{e}(t)| < \theta$, hence

$$|e(t)| \leq \theta t < \delta/2.$$

This rules out a first exit from the additive tube. For section trajectories,

$$w = q(p(t, \zeta) + e(t)), \quad h = p(t, \zeta) - p(0, \zeta) + e(t), \quad |e(t)| \leq \theta.$$

Since J is compact, these formulas confine that family to a compact set as well. All solutions therefore extend through time one, and the section estimate gives

$$|h(1) - (p(1, \zeta) - p(0, \zeta))| < \epsilon/2.$$

Lemma 3.1 thus applies to both compact initial families.

Step 4: frozen complete flows and the two final estimates. Expand all polynomial potentials into monomials and apply Lemma 3.1 to their augmented fields (3), with error less than $\epsilon/2$ in the displacement component. Since $q(z) \in K$ for every $z \in E_K$, all the corresponding augmented initial data lie in the compact set $K \times \{0\}$. Identity (5) identifies the final displacement with $A_\zeta(z) - z$, so the uniform estimate in the augmented coordinates gives (6) on the entire, possibly unbounded set E_K . For the section initial data, the same identity yields (7). $\square$

4 Entire sprays

This section closes the proof of Theorem 1.1. The first lemma upgrades a continuous real-time deformation to a path that is entire in the complex time variable, as required by Lemma 3.2. A fixed Fatou–Bieberbach model then turns the transported section into a normalized entire spray.

Lemma 4.1. *Suppose $Y \subset \mathbb{C}^n$ is open and path connected, $U \subset \mathbb{C}^m$ is convex, $L \Subset U$ is compact and convex, and $f : U \rightarrow Y$ is holomorphic. For every $p_* \in Y$ there exists a holomorphic map $p : \mathbb{C} \times U \rightarrow \mathbb{C}^n$ such that*

$$p(0, \zeta) = f(\zeta), \quad p(1, \zeta) = p_*, \quad p(t, \zeta) \in Y \quad (t \in [0, 1], \zeta \in L).$$

Proof. If L is empty, the affine path $p(t, \zeta) = (1 - t)f(\zeta) + tp_*$ will do, so assume that L is nonempty and choose $\zeta_0 \in L$. Contracting the source from ζ to ζ_0 inside U and then following a fixed path in Y from $f(\zeta_0)$ to p_* yields a continuous map $Q : [0, 1] \times U \rightarrow \mathbb{C}^n$, holomorphic in ζ at each fixed time, with $Q(0, \zeta) = f(\zeta)$ and $Q(1, \zeta) = p_*$. Its image on the compact set $[0, 1] \times L$ is compact in Y , so we may choose $d_Y > 0$ such that the d_Y -neighborhood of this image is contained in Y . The Bernstein polynomials

$$p_N(s, \zeta) = \sum_{j=0}^N Q(j/N, \zeta) \binom{N}{j} s^j (1-s)^{N-j}$$

are holomorphic on $\mathbb{C} \times U$ and preserve the endpoints exactly. To estimate them on $[0, 1] \times L$, write $M_Q = \sup_{[0, 1] \times L} |Q|$ and choose a modulus of continuity ω for Q in the time variable, uniform in $\zeta \in L$; such a modulus exists because Q is continuous on the compact product $[0, 1] \times L$. For every $a > 0$,

$$\sup_{[0, 1] \times L} |p_N - Q| \leq \omega(a) + \frac{M_Q}{2Na^2},$$

since the mass of the Bernstein kernel outside $|s - j/N| > a$ is at most $1/(4Na^2)$, its second moment being $s(1-s)/N \leq 1/(4N)$. Choosing a with $\omega(a) < d_Y/2$ and then N so large that $M_Q/(2Na^2) < d_Y/2$ gives $p_N([0, 1] \times L) \subset Y$. $\square$

For $M \in \mathbb{R}$, write

$$E_M^{(n)} = \{z \in \mathbb{C}^n : \operatorname{Im} z_j \leq M \text{ for every } j\}.$$

Appendix A constructs a biholomorphism

$$\Phi_2 : \mathbb{C}^2 \longrightarrow B_2 \subset \mathbb{C}^2 \setminus E_0^{(2)}, \quad \Phi_2(0) = (i, i), \quad D\Phi_2(0) = I_2.$$

The product map

$$\Phi(\xi) = (\Phi_2(\xi_1, \xi_2) + i(M+1)(1, 1), \xi_3, \dots, \xi_n) \tag{11}$$

is biholomorphic onto its image, satisfies $D\Phi(0) = I_n$, and is centered at

$$p_* = (i(M+2), i(M+2), 0, \dots, 0).$$

Its image avoids the intersection of the first two lower halfspaces $\{\operatorname{Im} z_1 \leq M+1, \operatorname{Im} z_2 \leq M+1\}$ and therefore avoids $E_{M+1}^{(n)}$.

Proof of Theorem 1.1. Nonemptiness and path connectedness follow from Lemma 2.1. If the source is empty, there is nothing to prove; if E_K is empty, then $Y_K = \mathbb{C}^n$, and affine sprays give the assertion. Suppose therefore that E_K is nonempty and set $M = \log \max\{1, \max_{w \in K} \max_j |w_j|\}$. Since $|e^{-iz_j}| = e^{\operatorname{Im} z_j}$,

$$E_K \subset E_M^{(n)}.$$

We use the fixed map (11), whose center p_* lies in Y_K .

Shrinking the original source if necessary, we may assume that it is a bounded convex open set U , and we choose a compact convex L with

$$L_0 \subset \operatorname{int} L \subset L \Subset U \Subset U_0.$$

By Lemma 4.1 there is a holomorphic time path from f to p_* that avoids E_K for $t \in [0, 1]$ and $\zeta \in L$. Applying Lemma 3.2 with $\epsilon < 1/4$, followed by the correction of the center by the holomorphic translation

$$\tilde{A}_\zeta(z) = A_\zeta(z) + p_* - A_\zeta(f(\zeta)),$$

we obtain $\tilde{A}_\zeta(f(\zeta)) = p_*$ together with

$$\sup_{z \in E_K} |\tilde{A}_\zeta(z) - z| < 2\epsilon < 1/2 \quad (\zeta \in L).$$

For $z \in E_K$ and every j ,

$$\operatorname{Im} \tilde{A}_{\zeta,j}(z) \leq \operatorname{Im} z_j + |\tilde{A}_{\zeta,j}(z) - z_j| < M + 2\epsilon < M + 1.$$

Hence $\tilde{A}_\zeta(E_K) \subset E_{M+1}^{(n)}$. On $V = \operatorname{int} L$, consider

$$C_\zeta = D\tilde{A}_\zeta(f(\zeta)), \quad F(\zeta, \xi) = \tilde{A}_\zeta^{-1}(\Phi(C_\zeta \xi)).$$

The matrices C_ζ are holomorphic and invertible, and the inverse automorphism family is jointly holomorphic, so F is holomorphic on $V \times \mathbb{C}^n$. Its image avoids E_K : indeed, Φ avoids $E_{M+1}^{(n)}$, while $\tilde{A}_\zeta(E_K)$ lies inside that set. Moreover

$$F(\zeta, 0) = f(\zeta), \quad D_\xi F(\zeta, 0) = C_\zeta^{-1} I_n C_\zeta = I_n.$$

Every fiber map is a composition of biholomorphisms onto their images, so the normalization $D_\xi F(\zeta, 0) = I_n$ makes F a dominating spray, and Kusakabe's convex spray criterion [11, Theorem 2.2] gives the Oka property of Y_K .

Since $q : Y_K \rightarrow X_K$ is a covering, Lemma 2.4 carries the Oka property over to X_K as well. $\square$

5 Inversion charts and slab crosses

The main device in this section is one-dimensional: translated inversion turns the complement of an arbitrary connected planar domain into a compact polynomially convex set. Applied coordinatewise, this places each punctured slab chart in the setting of Theorem 1.1; Kusakabe's localization theorem then assembles finitely many such charts.

Lemma 5.1. *For a nonempty connected planar domain U and $a \in U$, denote by J_a the Möbius transformation $z \mapsto 1/(z - a)$ of $\mathbb{P}^1$. Then*

$$K_a = J_a(\mathbb{P}^1 \setminus U) = \{0\} \cup \{1/(x - a) : x \in \mathbb{C} \setminus U\}$$

is a compact polynomially convex subset of $\mathbb{C}$ containing 0, and

$$\mathbb{C} \setminus K_a = J_a(U \setminus \{a\}), \quad w \in K_a \iff a + 1/w \notin U \quad (w \neq 0). \quad (12)$$

Proof. The compact set $\mathbb{P}^1 \setminus U$ does not contain the pole a , so its image is compact and avoids infinity; since $\infty \notin U$, that image contains $J_a(\infty) = 0$. Bijectivity gives the two identities in (12). The punctured planar domain $U \setminus \{a\}$ is path connected: a polygonal path that passes through a can be rerouted around a by a small circle inside U . Hence $\mathbb{C} \setminus K_a$ is connected. Since the planar polynomial hull consists exactly of K_a together with the bounded components of its complement, K_a is polynomially convex. For $U = \mathbb{C}$ one has $K_a = \{0\}$. $\square$

Thus connectedness of U is used at one precise point: it makes the complement of the inverted compact set connected, which in one variable is exactly the condition needed for polynomial convexity.

Proof of Theorem 1.2. Write $\mathcal{S} = \mathcal{S}(U_1, \dots, U_n)$ and fix $a_j \in U_j$. Consider

$$\mathcal{S}_a = \mathcal{S} \setminus \bigcup_{j=1}^n \{z_j = a_j\}, \quad K = \prod_{j=1}^n J_{a_j}(\mathbb{P}^1 \setminus U_j).$$

By Lemma 5.1 each compact factor is polynomially convex, and hence so is their product: a point outside the product has a coordinate outside the corresponding factor, and a one-variable polynomial in that coordinate separates it from the product. Coordinatewise inversion is a biholomorphism of the ambient chart with the coordinate hyperplanes $z_j = a_j$ removed onto $(\mathbb{C}^*)^n$, and (12) gives the identity

$$(J_{a_1}, \dots, J_{a_n})(\mathcal{S}_a) = (\mathbb{C}^*)^n \setminus (K \cap (\mathbb{C}^*)^n) = X_K.$$

Theorem 1.1 therefore makes $\mathcal{S}_a$ Oka.

We now choose two distinct points $a_{j,0}, a_{j,1} \in U_j$ for each j . The resulting 2^n charts cover $\mathcal{S}$, since for every z one may choose, independently in each coordinate, an a_{j,ε_j} different from z_j . Each chart is Zariski open in $\mathcal{S}$: its complement in $\mathcal{S}$ is the zero set of $\prod_j (z_j - a_{j,\varepsilon_j})$, hence a closed complex subvariety of $\mathcal{S}$. The manifold $\mathcal{S}$ is connected, its n connected open slabs having the nonempty common intersection $\prod_j U_j$, so $\mathcal{S}$ is Oka by Kusakabe's localization theorem [11, Theorem 1.4]. $\square$

When $n = 1$, the corresponding domain $\Omega_1 = \Delta$ is not Oka.

5.1 Removing finitely many points from a slab cross

Finite point removal illustrates a useful stability of the method. In an inversion chart, deleting finitely many points amounts to adjoining finitely many points to the compact polynomially convex obstacle. The elementary lemma below preserves polynomial convexity under that operation.

Corollary 5.2. *If $n \geq 2$, $U_1, \dots, U_n \subset \mathbb{C}$ are nonempty connected open sets, and $P \subset \mathcal{S}(U_1, \dots, U_n)$ is finite, then $\mathcal{S}(U_1, \dots, U_n) \setminus P$ is Oka.*

Lemma 5.3. *If $K \subset \mathbb{C}^n$ is compact and polynomially convex and $Q \subset \mathbb{C}^n$ is finite, then $K \cup Q$ is polynomially convex.*

Proof. Fix $x \notin K \cup Q$. Choose affine linear polynomials ℓ_q with $\ell_q(q) = 0$ and $\ell_q(x) \neq 0$, and set $h = \prod_{q \in Q} \ell_q$, with $h = 1$ if $Q = \emptyset$. If $K = \emptyset$, h separates x from Q . Otherwise choose a polynomial p with $|p(x)| > \sup_K |p|$. Since

$$\frac{\sup_K |hp^N|}{|h(x)p(x)^N|} \leq \frac{\sup_K |h|}{|h(x)|} \left(\frac{\sup_K |p|}{|p(x)|} \right)^N \longrightarrow 0,$$

the polynomial hp^N , which vanishes on Q , separates x from $K \cup Q$ for large N . $\square$

Proof of Corollary 5.2. Write $\mathcal{S} = \mathcal{S}(U_1, \dots, U_n)$. Use the 2^n inversion charts $\mathcal{S}_a$ and the compact polynomially convex sets K from the proof of Theorem 1.2. In each chart let $I_a = (J_{a_1}, \dots, J_{a_n})$ and $Q_a = I_a(P \cap \mathcal{S}_a)$. Then Q_a is finite, is disjoint from K , and

$$I_a(\mathcal{S}_a \setminus P) = (\mathbb{C}^*)^n \setminus (K \cup Q_a) = X_{K \cup Q_a}.$$

Lemma 5.3 and Theorem 1.1 show that every $\mathcal{S}_a \setminus P$ is Oka. These sets form an analytic-Zariski open cover of $\mathcal{S} \setminus P$. Since deleting finitely many points does not disconnect the connected open manifold $\mathcal{S}$, Kusakabe's localization theorem [11, Theorem 1.4] applies again and makes $\mathcal{S} \setminus P$ Oka. $\square$

Alarcón and Forstnerič [1, Corollary 2.9] also established a finite-removal result in the Oka-1 category. For punctured slab crosses, Corollary 5.2 gives the full Oka property.

6 Periodic Euclidean balls

Coordinatewise inversion turns the periodic ball obstacle into a relatively compact set whose closure meets the coordinate hyperplanes; Theorem 1.1 applies once that closure is known to be polynomially convex.

6.1 Polynomially convex compactification

Throughout this section the lattices and r satisfy the hypotheses of Theorem 1.3. For each j , consider the planar obstacle and its complement

$$A_j = \bigcup_{\lambda \in \Lambda_j} \overline{D}(\lambda, r), \quad U_j = \mathbb{C} \setminus A_j.$$

Here $\overline{D}(\lambda, r)$ denotes a closed planar disc. The discs are locally finite and have mutual distance at least $\ell_j - 2r > 0$, so A_j is closed. The annulus $r < |z| < \ell_j - r$ lies in U_j , and in particular U_j is nonempty.

To see that U_j is path connected, take two points of U_j . Since their distances from the closed set A_j are positive, a sufficiently small increase of the radius from r to some $s > r$ with $2s < \ell_j$ leaves both points outside every enlarged disc. A line segment joining the two points meets only finitely many of these discs; travelling around the boundary circle of each enlarged disc in place of the portion of the segment inside it gives a path in U_j : the enlarged discs are pairwise disjoint, and their boundary circles avoid all the original discs of radius r .

For $a = (a_1, \dots, a_n) \in \prod_j U_j$, the map

$$I_a : Z_a = \prod_{j=1}^n (\mathbb{C} \setminus \{a_j\}) \longrightarrow (\mathbb{C}^*)^n, \quad I_a(z) = \left(\frac{1}{z_1 - a_1}, \dots, \frac{1}{z_n - a_n} \right)$$

is a biholomorphism. The closed, locally finite union

$$E = \bigcup_{\lambda \in \Lambda} \overline{B}(\lambda, r)$$

is contained in Z_a , because its j th coordinate projection is A_j .

Lemma 6.1. *For every $a \in \prod_j U_j$, the closure $K = \overline{I_a(E)}$ in $\mathbb{C}^n$ is compact and polynomially convex, and*

$$K \cap (\mathbb{C}^*)^n = I_a(E). \quad (13)$$

Proof. The argument has four stages. First, the inverted obstacle is placed in a compact polynomially convex product envelope. Second, all strata on the coordinate hyperplanes are identified, together with the projections needed later. Third, each stratum with prescribed nonzero coordinates is shown to be polynomially convex. Finally, approximate component indicators localize a point of the polynomial hull to one stratum, where the preceding identities force it back into K .

Step 1: a compact product envelope. For $\lambda \in \Lambda_j$, write

$$D_{j,\lambda} = \left\{ \frac{1}{z - a_j} : |z - \lambda| \leq r \right\}, \quad P_j = \{0\} \cup \bigcup_{\lambda \in \Lambda_j} D_{j,\lambda}, \quad P = \prod_{j=1}^n P_j.$$

Since $a_j \notin \overline{D}(\lambda, r)$, its image $D_{j,\lambda}$ under the Möbius map $z \mapsto 1/(z - a_j)$ is a closed Euclidean disc not containing zero, and these discs are pairwise disjoint. Moreover,

$$\sup_{w \in D_{j,\lambda}} |w| \leq \frac{1}{|\lambda - a_j| - r} \longrightarrow 0 \quad (|\lambda| \longrightarrow \infty).$$

Since $P_j = J_{a_j}(\mathbb{P}^1 \setminus U_j)$, Lemma 5.1 shows that P_j is compact and polynomially convex.

Each disc $D_{j,\lambda}$ is separated from $P_j \setminus D_{j,\lambda}$ by a positive distance. Indeed, its distance from zero is positive, so all but finitely many of the other discs lie in a sufficiently small neighborhood of zero, while the remaining finitely many compact discs are disjoint from $D_{j,\lambda}$ and hence at positive distance from it. In particular, $D_{j,\lambda}$ is an isolated component of P_j .

The product P is polynomially convex by the coordinate separation argument used in the proof of Theorem 1.2.

Since $I_a(E) \subset P$ with P compact and polynomially convex, K is compact and $\widehat{K} \subset \widehat{P} = P$, where $\widehat{K}$ denotes the polynomial hull. If $I_a(z^\nu) \rightarrow w \in (\mathbb{C}^*)^n$ with $z^\nu \in E$, then

$$z^\nu \longrightarrow a + 1/w = \left(a_j + \frac{1}{w_j} \right)_{j=1}^n.$$

Closedness of E proves (13).

Step 2: boundary strata and their projections. For a nonempty set $I \subset \{1, \dots, n\}$ and a tuple $\lambda_I \in \prod_{i \in I} \Lambda_i$, write

$$B_{I,\lambda_I} = \left\{ z_I \in \mathbb{C}^I : \sum_{i \in I} |z_i - \lambda_i|^2 \leq r^2 \right\}, \quad Q_{I,\lambda_I} = I_{a,I}(B_{I,\lambda_I}),$$

where $I_{a,I}$ denotes the coordinatewise inversion restricted to the coordinates in I . Every coordinate of every point of Q_{I,λ_I} is nonzero.

For $w \in P \setminus \{0\}$, take $I = \{i : w_i \neq 0\}$ and denote by λ_i the unique label with $w_i \in D_{i,\lambda_i}$. Then

$$w \in K \iff w_I \in Q_{I,\lambda_I}. \quad (14)$$

For the forward implication, choose $z^\nu \in \overline{B}(\lambda^\nu, r)$ with $I_a(z^\nu) \rightarrow w$. Isolation of D_{i,λ_i} forces $\lambda_i^\nu = \lambda_i$ for all sufficiently large ν and for each $i \in I$, and since I is finite these identities hold simultaneously for large ν . Passing to the limit in

$$\sum_{i \in I} |z_i^\nu - \lambda_i|^2 \leq r^2$$

gives $a_I + 1/w_I \in B_{I,\lambda_I}$.

Conversely, suppose that $w_I \in Q_{I,\lambda_I}$ and keep $z_I = a_I + 1/w_I$ fixed. For each $j \notin I$, choose $\lambda_j^\nu \in \Lambda_j$ with $|\lambda_j^\nu| \rightarrow \infty$ and set $z_j^\nu = \lambda_j^\nu$. The center $(\lambda_I, \lambda_{I^c}^\nu)$ belongs to the full product lattice, and

$$|(z_I, z_{I^c}^\nu) - (\lambda_I, \lambda_{I^c}^\nu)|^2 = \sum_{i \in I} |z_i - \lambda_i|^2 \leq r^2.$$

Thus $(z_I, z_{I^c}^\nu) \in E$, and its inversion tends to w . When I is the full index set the constant sequence suffices, and choosing all coordinates to be lattice centers tending to infinity likewise shows that $0 \in K$.

Besides (14) we shall need a projection identity. For every nonempty I and every tuple λ_I , the compact subset

$$K_{I,\lambda_I} = \{w \in K : w_i \in D_{i,\lambda_i} \text{ for every } i \in I\}$$

satisfies

$$\text{pr}_I(K_{I,\lambda_I}) = Q_{I,\lambda_I}. \quad (15)$$

Indeed, for any $w \in K_{I,\lambda_I}$ the stabilization and limiting argument above, applied in the coordinates of I , gives $w_I \in Q_{I,\lambda_I}$ whatever the remaining coordinates are. Conversely, if $q \in Q_{I,\lambda_I}$, then by (14) the point with coordinates q on I and zero on I^c belongs to K , and since $q_i \in D_{i,\lambda_i}$ for every $i \in I$, this point lies in K_{I,λ_I} . Both inclusions in (15) follow.

Step 3: polynomial convexity of each stratum. The set Q_{I,λ_I} is contained in the polynomially convex product of discs

$$D_I = \prod_{i \in I} D_{i,\lambda_i} \subset (\mathbb{C}^*)^I,$$

so a point outside D_I is already separated from Q_{I,λ_I} by a polynomial. Suppose then that $p \in D_I \setminus Q_{I,\lambda_I}$. Then $a_I + 1/p$ lies outside the compact convex ball B_{I,λ_I} , and since a compact convex set is polynomially convex, some polynomial q satisfies

$$|q(a_I + 1/p)| > \sup_{z_I \in B_{I,\lambda_I}} |q(z_I)|.$$

The function $h(w_I) = q(a_I + 1/w_I)$ is holomorphic on a neighborhood of D_I , because D_I avoids every coordinate hyperplane. Oka–Weil approximation on D_I therefore produces polynomial approximants to h , uniformly on D_I , and an error below a third of the gap in the inequality above preserves the separation of p from Q_{I,λ_I} . Hence Q_{I,λ_I} is polynomially convex.

Step 4: localization of the polynomial hull. Fix $p \in \widehat{K} \subset P$. The case $p = 0$ was settled above, so assume $p \neq 0$. Write $I = \{i : p_i \neq 0\}$ for the set of its nonzero coordinates, and denote by λ_i the unique label with $p_i \in D_{i,\lambda_i}$. For $i \in I$, the function h_i on P_i that equals 1 on D_{i,λ_i} and 0 on its complement extends holomorphically to a neighborhood of P_i : it suffices to take disjoint open neighborhoods of these two compact sets and to prescribe the corresponding constant values on them. Oka–Weil approximation then supplies one-variable polynomials $h_{i,\nu}$ converging uniformly to h_i on P_i , so the polynomials

$$H_\nu(w) = \prod_{i \in I} h_{i,\nu}(w_i)$$

converge uniformly on P to $h(w) = \prod_{i \in I} h_i(w_i)$. This limit equals 1 at p , equals 1 on K_{I,λ_I} , and vanishes on $K \setminus K_{I,\lambda_I}$.

For any polynomial Q in the coordinates indexed by I , the hull inequality gives

$$|H_\nu(p)Q(p_I)| \leq \sup_{w \in K} |H_\nu(w)Q(w_I)|.$$

Uniform convergence on the compact set P , followed by (15), yields

$$|Q(p_I)| \leq \sup_{w \in K_{I,\lambda_I}} |Q(w_I)| = \sup_{q \in Q_{I,\lambda_I}} |Q(q)|.$$

Since Q_{I,λ_I} is polynomially convex, the inequality above gives $p_I \in Q_{I,\lambda_I}$, and (14) then gives $p \in K$. Hence $\widehat{K} = K$. $\square$

The lemma is the only application-specific convexity ingredient in Theorem 1.3. The functions H_ν in the last step do not further approximate the obstacle; they localize the hull inequality to the unique block K_{I,λ_I} selected by the nonzero coordinates of p .

6.2 Oka localization

Proof of Theorem 1.3. For every $a \in \prod_j U_j$, Lemma 6.1 provides a compact polynomially convex set $K_a = \overline{I_a(E)}$ together with a biholomorphism

$$I_a : \Omega \cap Z_a \longrightarrow (\mathbb{C}^*)^n \setminus K_a.$$

Theorem 1.1 therefore applies to each $\Omega \cap Z_a$. Choose two distinct points $a_{j,0}, a_{j,1} \in U_j$ for each j . As in the proof of Theorem 1.2, the 2^n sets

$$W_\varepsilon = \Omega \setminus \bigcup_{j=1}^n \{z_j = a_{j,\varepsilon_j}\}, \quad \varepsilon \in \{0,1\}^n,$$

form an analytic-Zariski open cover of Ω .

The nonempty product $\prod_j U_j$ lies in Ω . Given two points of Ω , their positive distances from the closed set E allow us to enlarge the radius to some s with $r < s < \ell/2$ while keeping both points outside all the enlarged balls. A segment joining the two points meets only finitely many such balls, and replacing each portion inside one of them by a path on its boundary sphere yields a path in Ω , since these spheres are path connected and avoid every original ball. The analytic-Zariski open cover above is therefore a cover by Oka domains, and Kusakabe's localization theorem [11, Theorem 1.4] makes Ω Oka. $\square$

7 Products of elliptic curves

7.1 Product holes

This theorem is the quotient form of Theorem 1.2. Under the product universal covering, the full inverse image of the product-hole domain is exactly a slab cross; connectedness of the planar lifts is the precise topological hypothesis that makes the reduction work.

Theorem 7.1. *Suppose $n \geq 2$. For $1 \leq j \leq n$, suppose $E_j = \mathbb{C}/\Lambda_j$ is an elliptic curve with quotient map $\pi_j : \mathbb{C} \rightarrow E_j$, and $D_j \subset E_j$ is a nonempty connected open set whose full inverse image $U_j = \pi_j^{-1}(D_j)$ is connected. Then*

$$X = \bigcup_{j=1}^n (E_1 \times \cdots \times E_{j-1} \times D_j \times E_{j+1} \times \cdots \times E_n)$$

is Oka. Equivalently, writing $K_j = E_j \setminus D_j$, the domain

$$(E_1 \times \cdots \times E_n) \setminus (K_1 \times \cdots \times K_n)$$

is Oka.

Proof. The product universal covering $\pi = \pi_1 \times \cdots \times \pi_n$ has full inverse image

$$\pi^{-1}(X) = \mathcal{S}(U_1, \dots, U_n).$$

Each U_j is a nonempty connected planar domain, so Theorem 1.2 makes the inverse image Oka. Its image X is connected, and the restriction of π is an unramified holomorphic covering onto X , so Lemma 2.4 applies to X . $\square$

The connected-lift hypothesis can be tested topologically: for a nonempty connected open set $D \subset E$, the full lift to $\mathbb{C}$ is connected if and only if

$$\pi_1(D) \longrightarrow \pi_1(E) = \Lambda$$

is surjective. Indeed, the components of the full lift are indexed by the cosets of the image subgroup in the deck group Λ , so the lift is connected exactly when the image is all of Λ .

Corollary 7.2. *Suppose $n \geq 2$ and $E_1, \dots, E_n$ are elliptic curves. If each $K_j \subset E_j$ is a closed embedded disc, then*

$$(E_1 \times \cdots \times E_n) \setminus (K_1 \times \cdots \times K_n)$$

is Oka. The same conclusion holds when each K_j is a finite union of pairwise disjoint closed embedded discs.

The point is that the connected-lift hypothesis in Theorem 7.1 becomes a concrete fundamental-group test for disc complements.

Proof of Corollary 7.2. Given two points of $D_j = E_j \setminus K_j$, the finitely many closed discs can be enlarged slightly to pairwise disjoint closed discs that avoid both points, and rerouting a path between the points along the enlarged boundary circles shows that D_j is connected.

Write $M_j \subset D_j$ for the connected compact surface obtained by removing the interiors of the enlarged discs. Filling in its boundary circles with discs recovers E_j , so van Kampen's theorem gives a surjection $\pi_1(M_j) \rightarrow \pi_1(E_j)$. The inclusions $M_j \subset D_j \subset E_j$ factor this surjection through $\pi_1(D_j)$, whose image is therefore all of $\pi_1(E_j)$. The covering-space criterion above now shows that each $U_j = \pi_j^{-1}(D_j)$ is connected, and Theorem 7.1 applies. $\square$

Theorem 7.1 also applies when each compact K_j lies in a coordinate disc and $E_j \setminus K_j$ is connected, since the two generators of $\pi_1(E_j)$ can then be represented outside a slightly larger disc. Containment in a disc without connectedness is insufficient for this argument, since a simple closed curve separates off an interior component.

7.2 A closed metric ball

For the flat product torus T and the lattices fixed in the Introduction, the full inverse image of a closed ball whose radius is below the injectivity radius is exactly the periodic obstacle of Theorem 1.3.

Proof of Corollary 1.4. By translation we may take $c = 0$, and denote the quotient map by $\pi : \mathbb{C}^n \rightarrow T$. Two points of $\overline{B}(0, r)$ differ by a vector of length at most $2r < \ell$, so their difference cannot be a nonzero lattice vector; the same argument applies to a slightly larger radius. Hence π embeds a neighborhood of $\overline{B}(0, r)$, and $\pi(\overline{B}(0, r)) = \overline{B}_T(0, r)$. Moreover,

$$\pi^{-1}(\overline{B}_T(0, r)) = \pi^{-1}(\pi(\overline{B}(0, r))) = \bigcup_{\lambda \in \Lambda} \overline{B}(\lambda, r) = E.$$

Thus π restricts to an unramified holomorphic covering from the Oka manifold Ω onto $T \setminus \overline{B}_T(0, r)$; by Lemma 2.4, $T \setminus \overline{B}_T(0, r)$ is Oka as well. $\square$

A An explicit two-dimensional fixed model

This appendix supplies the single fixed Fatou–Bieberbach model used in (11). Two explicit shears push the lower quadrant strictly downward while fixing an attracting point; the attracting basin then avoids the obstacle and is biholomorphic to $\mathbb{C}^2$.

Write $E_0^{(2)} = \{\operatorname{Im} z_1 \leq 0, \operatorname{Im} z_2 \leq 0\}$ and $p = (i, i)$. The polynomials

$$Q_1(\xi) = 3(\xi/e)^5(6 - 5\xi/e), \quad Q_2(\xi) = 3(\xi/e^4)(2 - \xi/e^4)$$

take the value 3 at e and e^4 , respectively, and their derivatives vanish at those points. Moreover

$$\begin{aligned}\sup_{|\xi| \leq 1} |Q_1(\xi)| &\leq 3e^{-5}(6 + 5e^{-1}) < \frac{1}{2}, \\ \sup_{|\xi| \leq e^{1/2}} |Q_2(\xi)| &\leq 3e^{-7/2}(2 + e^{-7/2}) < \frac{1}{2}.\end{aligned}$$

Consider the shears

$$\begin{aligned}A_1(z) &= (z_1 + iQ_1(e^{-iz_2}), z_2), & A_2(z) &= (z_1, z_2 + iQ_2(e^{-iz_1})), \\ \Psi(z) &= \frac{1}{2}(A_2 \circ A_1)(z) - i(1, 1).\end{aligned}$$

Each shear, and therefore Ψ , is an entire automorphism. The successive images of p under the two shears are $(4i, i)$ and $(4i, 4i)$, so $\Psi(p) = p$ and $D\Psi(p) = \frac{1}{2}I_2$. For $z \in E_0^{(2)}$, the first shear gives $\text{Im}(A_1 z)_1 < 1/2$ and leaves the second coordinate unchanged, while the second polynomial is evaluated at modulus at most $e^{1/2}$, so its output has $\text{Im}(A_2 A_1 z)_2 < 1/2$ without changing the first coordinate. Hence

$$\Psi(E_0^{(2)}) \subset \{\text{Im } z_1 < -3/4, \text{Im } z_2 < -3/4\} \subset E_0^{(2)}.$$

The attracting basin of p avoids this closed forward-invariant set: the iterates of a point of that set remain in it, whereas those of a basin point converge to p .

The eigenvalues of $D\Psi(p)$ are both $1/2$, so $|1/2|^2 < |1/2| < 1$. Rosay–Rudin’s attracting-basin theorem [14, Theorem 9.1] therefore gives a biholomorphism from the basin B_2 of p onto $\mathbb{C}^2$. Write $\psi(u) = \Psi(p + u) - p$. In these translated coordinates, the theorem’s map is $H = \lim_{N \rightarrow \infty} 2^N \psi^N$, with $H(0) = 0$ and $DH(0) = I_2$. Thus its inverse, translated back by p , gives

$$\Phi_2 : \mathbb{C}^2 \longrightarrow B_2 \subset \mathbb{C}^2 \setminus E_0^{(2)}, \quad \Phi_2(0) = p, \quad D\Phi_2(0) = I_2,$$

and this is the biholomorphism used in (11).

Funding

Partial support for S.-Y. Xie came from the National Key R&D Program of China through grants 2023YFA1010500 and 2021YFA1003100, from the National Natural Science Foundation of China through grants 12288201 and 12471081, and from the Xiaomi Young Talents Program.

AI use statement

The authors developed the mathematical ideas and arguments. AI assistance was used for manuscript preparation. Responsibility for reviewing and approving all content rests with the authors.

References

- [1] A. Alarcón and F. Forstnerič, *Oka-1 manifolds*, Math. Z. **311** (2025), article 33. doi:10.1007/s00209-025-03833-4.
- [2] E. Andersén and L. Lempert, *On the group of holomorphic automorphisms of $\mathbb{C}^n$* , Invent. Math. **110** (1992), 371–388. doi:10.1007/BF01231337.

- [3] Y.-H. Du, B. Guo, P.-C. Wang, and S.-Y. Xie, *Periodic transport and Oka complements in $\mathbb{C}^n$* , preprint, 2026. arXiv:2609.28089v2. Proposition 1.2 refers to the first version, *Periodic Hamiltonian approximation and Oka complements in $\mathbb{C}^2$* , arXiv:2609.28089v1.
- [4] F. Forstnerič, *Interpolation by holomorphic automorphisms and embeddings in $\mathbb{C}^n$* , J. Geom. Anal. **9** (1999), no. 1, 93–117. doi:10.1007/BF02923090.
- [5] F. Forstnerič, *Holomorphic flexibility of complex manifolds*, preprint, version 2, 16 March 2004. arXiv:math/0401439v2.
- [6] F. Forstnerič, *Oka manifolds: From Oka to Stein and back*, Ann. Fac. Sci. Toulouse Math. (6) **22** (2013), no. 4, 747–809. doi:10.5802/afst.1388.
- [7] F. Forstnerič, *Stein Manifolds and Holomorphic Mappings: The Homotopy Principle in Complex Analysis*, second edition, Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge, vol. 56, Springer, Cham, 2017. doi:10.1007/978-3-319-61058-0.
- [8] F. Forstnerič and F. Lárusson, *Survey of Oka theory*, New York J. Math. **17a** (2011), 11–38. Official journal page.
- [9] F. Forstnerič and E. Fornæss Wold, *Holomorphic families of Fatou–Bieberbach domains and applications to Oka manifolds*, Math. Res. Lett. **27** (2020), no. 6, 1697–1706. doi:10.4310/MRL.2020.v27.n6.a5.
- [10] F. Forstnerič and E. Fornæss Wold, *Oka domains in Euclidean spaces*, Int. Math. Res. Not. IMRN **2024** (2024), no. 3, 1801–1824. doi:10.1093/imrn/rnac347.
- [11] Y. Kusakabe, *Elliptic characterization and localization of Oka manifolds*, Indiana Univ. Math. J. **70** (2021), no. 3, 1039–1054. doi:10.1512/iumj.2021.70.8454. The numbered criteria refer to arXiv:1808.06290v1, Theorems 2.2 and 1.4.
- [12] Y. Kusakabe, *Oka properties of complements of holomorphically convex sets*, Ann. of Math. (2) **199** (2024), no. 2, 899–917. doi:10.4007/annals.2024.199.2.7. The question and corollary are cited with the numbering of arXiv:2005.08247v2.
- [13] F. Lárusson, *Eight lectures on Oka manifolds*, lecture notes, 2014. arXiv:1405.7212.
- [14] J.-P. Rosay and W. Rudin, *Holomorphic maps from $\mathbb{C}^n$ to $\mathbb{C}^n$* , Trans. Amer. Math. Soc. **310** (1988), no. 1, 47–86. doi:10.1090/S0002-9947-1988-0929658-4.
- [15] E. L. Stout, *Polynomial Convexity*, Progress in Mathematics, vol. 261, Birkhäuser Boston, Boston, MA, 2007. doi:10.1007/978-0-8176-4538-0.

(YUN-HENG DU) ACADEMY OF MATHEMATICS AND SYSTEMS SCIENCE, CHINESE ACADEMY OF SCIENCES, BEIJING 100190, CHINA.

Email address: duyunheng@amss.ac.cn

(BIN GUO) ACADEMY OF MATHEMATICS AND SYSTEMS SCIENCE, CHINESE ACADEMY OF SCIENCES, BEIJING 100190, CHINA.

Email address: guobin181@mails.ucas.ac.cn

(PENG-CHAO WANG) ACADEMY OF MATHEMATICS AND SYSTEMS SCIENCE, CHINESE ACADEMY OF SCIENCES, BEIJING 100190, CHINA.

Email address: wangpengchao26@mails.ucas.ac.cn

(SONG-YAN XIE) STATE KEY LABORATORY OF MATHEMATICAL SCIENCES, ACADEMY OF MATHEMATICS AND SYSTEMS SCIENCE, CHINESE ACADEMY OF SCIENCES, BEIJING 100190, CHINA; SCHOOL OF MATHEMATICAL SCIENCES, UNIVERSITY OF CHINESE ACADEMY OF SCIENCES, BEIJING 100049, CHINA.
Email address: `xiesongyan@amss.ac.cn`